

Q1 - 25 July - Shift 2

Light wave traveling in air along x-direction is given by $E_y = 540 \sin \pi \times 10^4 (x - ct) \text{ Vm}^{-1}$.

Then, the peak value of magnetic field of wave will be (Given $c = 3 \times 10^8 \text{ ms}^{-1}$)

- (A) $18 \times 10^{-7} \text{ T}$ (B) $54 \times 10^{-7} \text{ T}$
(C) $54 \times 10^{-8} \text{ T}$ (D) $18 \times 10^{-8} \text{ T}$

Space for your notes:

Q2 - 26 July - Shift 1

The magnetic field of a plane electromagnetic wave is given by

$$\vec{B} = 2 \times 10^{-8} \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t) \hat{j} \text{ T}$$

The amplitude of the electric field would be

- (A) 6 Vm^{-1} along x-axis
(B) 3 Vm^{-1} along z-axis
(C) 6 Vm^{-1} along z-axis
(D) $2 \times 10^{-8} \text{ Vm}^{-1}$ along z-axis

Space for your notes:

Q3 - 26 July - Shift 2

The oscillating magnetic field in a plane electromagnetic wave is given by $B_y = 5 \times 10^{-6} \sin 1000 \pi (5x - 4 \times 10^8 t) \text{ T}$. The amplitude of electric field will be :

- (A) $15 \times 10^2 \text{ Vm}^{-1}$ (B) $5 \times 10^{-6} \text{ Vm}^{-1}$
(C) $16 \times 10^{12} \text{ Vm}^{-1}$ (D) $4 \times 10^2 \text{ Vm}^{-1}$

Space for your notes:

Q4 - 27 July - Shift 2

Identify the correct statements from the following descriptions of various properties of electromagnetic waves.

- A. In a plane electromagnetic wave electric field and magnetic field must be perpendicular to each other and direction of propagation of wave should be along electric field or magnetic field.
- B. The energy in electromagnetic wave is divided equally between electric and magnetic fields.
- C. Both electric field and magnetic field are parallel to each other and perpendicular to the direction of propagation of wave.
- D. The electric field, magnetic field and direction of propagation of wave must be perpendicular to each other.
- E. The ratio of amplitude of magnetic field to the amplitude of electric field is equal to speed of light.

Choose the most appropriate answer from the options given below:

- (A) D only
- (B) B and D only
- (C) B, C and E only
- (D) A, B and E only

Space for your notes:

Q5 - 29 July - Shift 1

Match List – I with List – II

List – I	List - II
(a) UV rays	(i) Diagnostic tool in medicine
(b) X-rays	(ii) Water purification
(c) Microwave	(iii) Communication, Radar
(d) Infrared wave	(iv) Improving visibility in foggy days

Choose the correct answer from the options given below :

- (A) (a)–(iii), (b)–(ii), (c)–(i), (d)–(iv)
(B) (a)–(ii), (b)–(i), (c)–(iii), (d)–(iv)
(C) (a)–(ii), (b)–(iv), (c)–(iii), (d)–(i)
(D) (a)–(iii), (b)–(i), (c)–(ii), (d)–(iv)

Space for your notes:

Answer Key

Q1 (A)

Q2 (C)

Q3 (D)

Q4 (B)

Q5 (B)

#MathBoleTohMathonGo

Q1 (A)

$$E_y = 540 \sin \pi \times 10^4 (x - ct) \text{ Vm}^{-1}$$

$$E_0 = 540 \text{ Vm}^{-1}$$

$$B_0 = \frac{E_0}{c} = \frac{540}{3 \times 10^8} = 18 \times 10^{-7} \text{ T}$$

Q2 (C)

$$c = \frac{E_0}{B_0} \Rightarrow E_0 = cB_0$$

$$E_0 = (3 \times 10^8)(2 \times 10^{-8})$$

$$E_0 = 6 \text{ Vm}^{-1}$$

As, $\vec{B}$ = along y-axis

$\vec{v}$ = along negative x-axis

hence $\vec{E}_0$ = along z-axis

Q3 (D)

$$B_0 = 5 \times 10^{-6}$$

$$v = \text{Speed of wave} = \frac{4 \times 10^8}{5} = 8 \times 10^7 \left[\because v = \frac{w}{k} \right]$$

$$E_0 = vB_0 = 40 \times 10^1$$

$$= 4 \times 10^2 \text{ V/m}$$

#MathBoleTohMathonGo

Q4 (B)**ANS. (B)**

Second and fourth statements are correct.

Q5 (B)

- (a) uv rays – used for water purification
- (b) x-rays used for diagnosing fracture
- (c) Microwaves are used for mobile and radar communication
- (d) Infrared waves show less scattering therefore used in foggy days

(a – ii), (b – i), (c – iii), (d – iv)